

EXPERIENCE

- Software Developer Engineer II, Project 990 – Bloomington, IN

Jul 2025 – Present

 - Achieved 8x speedup in query response time over a 5M node & 50M relationship graph database by adding concurrent traversal start points, using hot path, introducing caching and tuning indexes and memory settings of the database
 - Implemented CI/CD pipelines with GitHub Actions to test & deploy .NET 5 microservices on cutting deploy time 80%.
 - Tech: .NET 5 (ASP.NET Core), Python, AWS, Gremlin, Docker, Kubernetes, GitHub Actions, REST, JSON
- Senior Software Engineer, App Orchid – Hyderabad, India

Apr 2022 – Aug 2023

 - Led 3 engineers in re-architecture of scalable ML data ingestion engine generating \$1M ARR and was assigned owner
 - Implemented Java REST APIs with Spring Boot, integrating auth, input validation, and JSON contracts.
 - Integrated microservices with PostgreSQL & DynamoDB using AWS EC2, S3 improving scalability & reliability.
 - Automated build processes using Azure Docker and Kubernetes service, implementing CI/CD pipeline to ensure quality
 - Tech: Java, Spring Boot, PostgreSQL, DynamoDB, AWS, Docker, Kubernetes, React, SQL, NoSQL, Restful API
- Software Engineer, HDFC Bank – Bengaluru, India

Jul 2020 – Apr 2022

 - Built a scalable risk classification ML pipeline for 50K+ audio calls, deployed as cloud-hosted serverless services
 - Designed batch ETL workflows in Apache Airflow with fan-out/fan-in parallel steps and automatic retries on failure
 - Constructed database schemas & replaced LINQ with T-SQL stored procedures resulting in 25% increase in performance
 - Tech: Python, Apache Airflow, Docker, Kubernetes, PostgreSQL, Redis, Apache Kafka, Elasticsearch, BERT
- Generative AI Intern, Center for Unified Biometrics and Sensors – Buffalo, NY

May 2024 – Aug 2024

 - Built RAG pipeline with Selenium scraping and integrated RAG & VLM on robot Nao for multimodal conversations
 - Tech: FAISS, LangChain, GPT-4, Selenium, Django, Flask, Python, chromaDB,
- Graduate Research Assistant, Prof. Prateek Sharma – Indiana University

Aug 2024 – May 2025

 - Designed and implemented high-performance distributed inference services in C++ and CUDA for LLM workloads, optimizing memory layout, concurrency, and GPU utilization to reduce latency by 20x
 - Developed low-latency communication pipelines using gRPC and isolated testing environments using Docker
 - Optimized system performance via multi-threading, stream concurrency and memory management
 - Publication:** MQFQ-Sticky — Fair Queueing for Serverless GPU Functions

PROJECTS

- Online Coding Platform | Website

Open Source Project

 - Built a full-stack app with React frontend and FastAPI backend, fully containerized and deployed on Azure.
 - Leveraged Azure Container Apps, Static Web Apps, and Blob Storage for scalable deployment.
 - Automated provisioning and redeployment using Terraform and CI/CD pipelines.
 - Integrated an in-browser VS Code IDE for secure code execution in isolated containers.
- Library Management Web App | Website

Software Engineering I

 - Developed a MERN stack web app with FastAPI services for catalogue, booking, and role-based access.
 - Implemented CI/CD with GitHub Actions and automated testing for continuous delivery.
 - Collaborated in Agile/Scrum sprints using Jira for task tracking and sprint planning.

SKILLS

Languages	C/C++, Java, Python, JavaScript, TypeScript, SQL
Technology	.NET Core, React, Node, Spring Boot, REST API, Flask, FastAPI, Django, Git, Jira
Data & Tools	AWS, EC2, DynamoDB, S3, SQS, Lambda, Elasticsearch, Spark, Docker, Splunk, Kafka, RabbitMQ

EDUCATION

- M.S. in Computer Science, **Indiana University**, Bloomington, IN | GPA: 3.8/4.0
- Aug 2023 – May 2025
- B.Tech in Aerospace, Minor in Computer Science, **Indian Institute of Technology, Bombay**
- Jul 2016 – Jul 2020